



Automated Backup System using AWS Lambda

"A serverless AWS solution that automates Amazon EBS volume backups using AWS Lambda, ensuring secure, scheduled, and cost-effective data protection."

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AGENDA

- Problem Statement
 - Solution Overview
 - Key Features
 - AWS Services used
 - Architecture
 - Future Work
 - Conclusion
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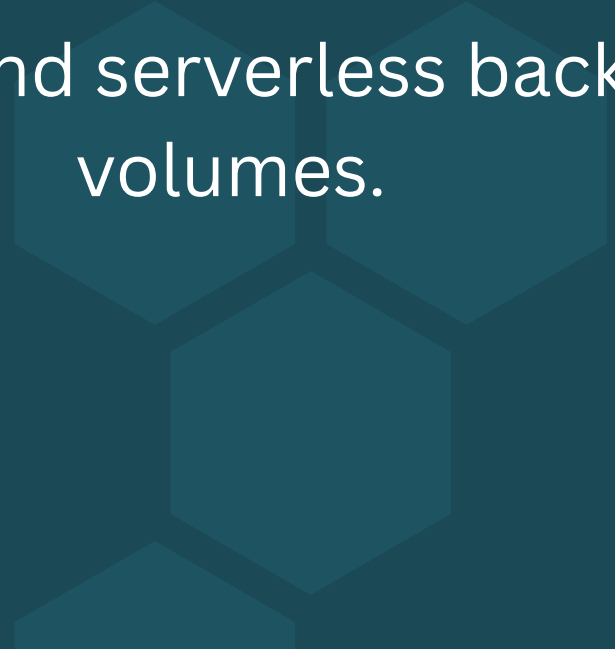


Problem Statement

- Manual backup of cloud resources is time-consuming.
- Risk of data loss due to accidental deletion or failures.
- Storage costs increase without automated cleanup.
- Traditional backup systems require continuous administration.

Need:

A secure, automated, scalable, and serverless backup solution for Amazon EBS volumes.

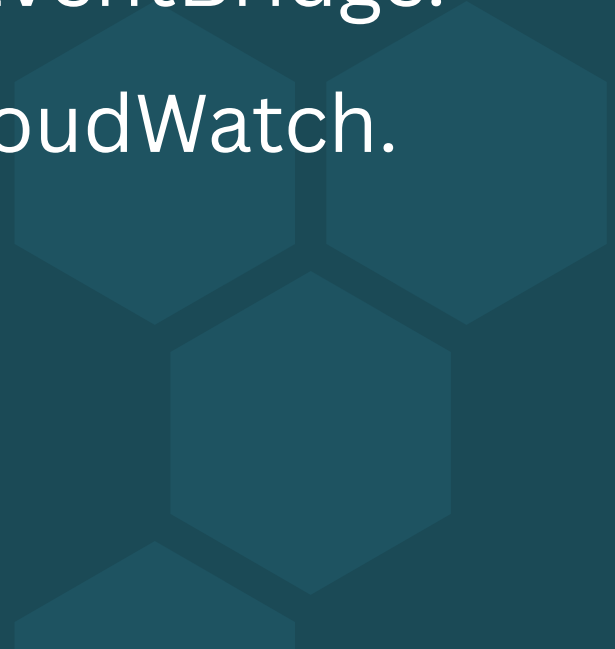




Solution Overview

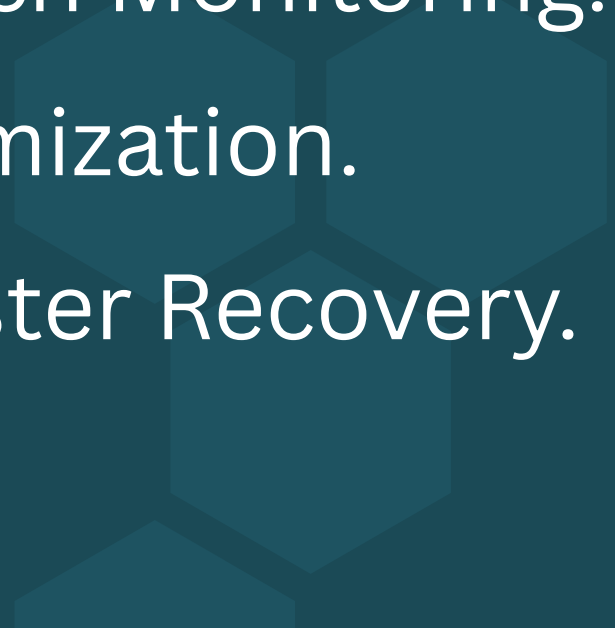
The system automates Amazon EBS volume backups using AWS Lambda.

Workflow:

- Identify EBS volumes tagged Backup=true.
 - Create snapshots automatically.
 - Delete old snapshots after retention period.
 - Schedule execution using EventBridge.
 - Monitor execution using CloudWatch.
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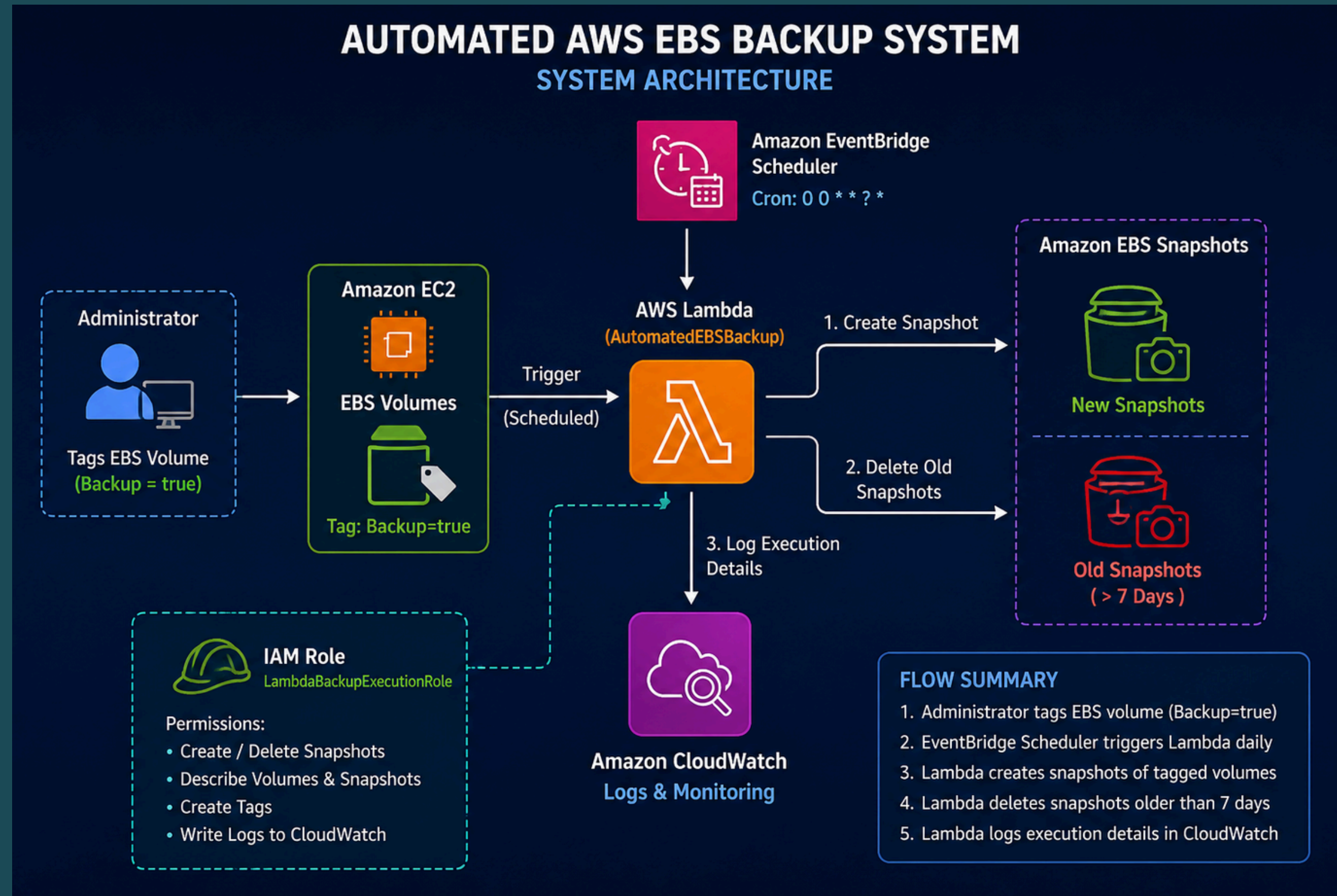
Key Features

- Automated EBS Snapshot Creation.
 - Tag-based Volume Selection.
 - Snapshot Retention Policy.
 - Daily Scheduled Backups.
 - Serverless Architecture.
 - IAM-based Secure Access.
 - CloudWatch Monitoring.
 - Cost Optimization.
 - Easy Disaster Recovery.
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AWS Services Used

AWS Service	Purpose
Amazon EC2	Hosts application instances
Amazon EBS	Storage volumes
AWS Lambda	Backup automation
Amazon EventBridge	Scheduled execution
AWS IAM	Security & permissions
Amazon CloudWatch	Monitoring & logging
Amazon EBS Snapshots	Backup storage

System Architecture





Future Work

- Cross-region Backup Replication.
 - Amazon SNS Email Notifications.
 - AWS Backup Integration.
 - Backup Monitoring Dashboard.
 - Lifecycle Policies.
 - Compliance Reporting.
 - API-based Backup Management.
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Conclusion

- Successfully automated Amazon EBS backups.
- Reduced manual effort through serverless automation.
- Improved security using IAM.
- Optimized storage using retention policies.
- Achieved reliable, scalable, and cost-effective cloud backup management.



THANK YOU